



FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	1.6821	Longitude	24.0244
Zone size (km)	10	Analysis date	2026-09-30

80

FOREST INTEGRITY SCORE / 100 Healthy Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.827	Excellent
NDVI Trend 3 years (Sentinel-2)	+0.029	Stable
10-year trend (Landsat 8/9)	+0.463	Improving
40-year trend (Landsat 1984)	+0.534	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+1.34% / year	Amelioration
NDVI change 5 years	+0.034	Stable
Leaf type	Mixed (Coniferous/Broadleaf)	Mixed needles and flat leaves
Developmental stage	Old-growth forest (>120 years)	Ancient forest, high biodiversity
Above-ground biomass	~174.1 t/ha	High biomass (150-300 t/ha)
Canopy Height (GEDI+S2 10m)	16m max / 8.9m median	Tree height from LiDAR (ISS) fused with Sentinel-2 at 10m resolution
Carbon Assessment (IPCC)	2,971,954 t CO ₂	CO ₂ sequestered in this forest zone — IPCC methodology (AGB+BGB × 0.47 × 3.67)
CO ₂ per hectare	378.4 t CO ₂ /ha	Carbon density per hectare
Zone area	7,854 ha	Analysis zone area in hectares
Carbon credit value	EUR 44,579,310	Estimated value at EUR 15/tonne CO ₂ (voluntary carbon market)
REDD+ MRV Monitoring	Compliant	This report provides satellite-based Measurement, Reporting and Verification (MRV) data for the analyzed forest zone, consistent with REDD+ monitoring requirements. CO ₂ sequestration calculated using IPCC Tier 1 methodology from NASA GEDI LiDAR biomass data.

Methodology	IPCC 2006/2019 — Tier 1	Above-ground biomass measured via NASA GEDI LiDAR (25m, ISS). Carbon fraction: 0.47. CO2 conversion factor: 3.67. Below-ground biomass ratio: 0.26 (IPCC root-to-shoot). Source: IPCC Guidelines for National GHG Inventories (2019).
Carbon credit potential	EUR 3,714,942 / year	Estimated annual value based on maintained forest preservation. For official REDD+ credit issuance under VCS or Gold Standard, independent third-party verification is required.
GEDI Tree Cover	98.9%	Percentage of the zone covered by trees according to NASA GEDI
AI Forest Type	Dense Tropical Forest	High biodiversity, dense canopy
Deforestation Risk (AI)	Low	Low risk
Main cause (AI)	Healthy natural forest ecosystem in Central Africa (Chad region)	
Recovery signs (AI)	Excellent vegetation health with very high NDVI (0.839) and EVI (2.625) indicating robust photosynthetic activity	
AI Recommendation	Continue monitoring vegetation trends; maintain current conservation status as this area shows optimal forest health indicators.	
Sentinel-1 Radar (VV)	-7.326 dB	Dense forest (radar)
Forest cover	100%	Excellent
Fire pixels detected (FIRMS/year)	302 pixels	~76 events estimated
Overall trend	stable	
Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5 - NASA GEDI LiDAR	
Deforestation Alert	STABLE — No significant change	
Area change (3 years)	+0 ha	+0%
Annual rate (area/yr)	+0 ha/yr	Recovery
Area change (10 years)	+9802.3 ha	+5508.12%
Annual rate 10yr (area)	+980.2 ha/yr	Recovery

SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	25	25
NDVI trend (3 years)	15	25
Forest cover	25	25
Sentinel-1 Radar (VV)	15	15
Fire score (0=many fires, 10=none)	0	10
TOTAL	80	100

SPECIES CLIMATE SUITABILITY (2050)

Scenario: SSP2-4.5 (2050) — 5-model ensemble | Temperature: 24.7°C → 27.2°C (+2.5°C) | Precipitation: 1741 → 1683 mm/year (-3.4%)

Species	Family	Score 2050	Status	Temp. range
Asplenium africanum	Aspleniaceae	1	Suitable	20.8–29.0°C
Rinorea laurentii	Violaceae	1	Suitable	21.8–28.8°C
Pyrenacantha thomsoniana	Icacinaceae	1	Suitable	21.2–28.7°C
Pyrenacantha sylvestris	Icacinaceae	1	Suitable	21.4–28.7°C
Marantochloa purpurea	Marantaceae	0.97	Suitable	18.9–28.5°C
Ceratopteris cornuta	Pteridaceae	0.91	Suitable	-2.2–28.4°C
Microgramma mauritiana	Polypodiaceae	0.9	Suitable	9.1–29.0°C
Cyathula achyranthoides	Amaranthaceae	0.87	Suitable	-2.1–28.9°C
Murdannia simplex	Commelinaceae	0.8	Suitable	10.1–29.0°C
Pityrogramma calomelanos	Pteridaceae	0.79	Suitable	10.9–28.9°C
Oleandra distenta	Oleandraceae	0.71	Suitable	9.5–28.9°C
Pontederia crassipes	Pontederiaceae	0.7	Suitable	4.0–28.6°C
Lantana camara	Verbenaceae	0.7	Suitable	4.8–28.9°C
Nymphaea lotus	Nymphaeaceae	0.7	Suitable	0.7–29.0°C
Flagellaria guineensis	Flagellariaceae	0.62	Moderate risk	8.0–26.7°C

METHODOLOGY — Bioclimatic Envelope Assessment (BEA): Suitability scores are calculated by comparing projected 2050 temperature (NASA GDDP-CMIP6, 5-model ensemble: ACCESS-CM2, MIROC6, MPI-ESM1-2-HR, GFDL-ESM4, BCC-CSM2-MR, SSP2-4.5 scenario) with each species real thermal tolerance range, derived from global distribution analysis via GBIF (100 occurrences/species). Method: temperature estimated from latitude ($27 - |\text{lat}| \times 0.52^\circ\text{C}$), proportional penalties when 2050 temperature exceeds tolerated range. Thresholds: Suitable ≥ 0.70 | Moderate risk 0.40-0.69 | At risk < 0.40 . This BEA approach is consistent with full MaxEnt results for general trends (cf. published studies on Ceiba pentandra, SSP2-4.5, Frontiers in Forests 2026). Data sources: GBIF (CC BY 4.0) · WorldClim V1 · NASA GDDP-CMIP6 · ESA Sentinel-2.

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov